



Learn Citaville Institute - Brochure

Advanced Data Analysis using Python & SQL

About the Course Program

This 8–10 week program introduces advanced data analytics using **Python** and **SQL**. You'll learn programming for data wrangling, database queries, and predictive analysis for real-world problem-solving.

Delivery: Virtual (online, coding-focused with labs)

Certification Requirement: Passing coding assignments and project assessments.

Career Paths: Data Analyst, Data Engineer, Junior Data Scientist.

Program Duration: 8–10 weeks

Who Can Attend?

This program is designed for:

- Students or professionals with some programming background
- Analysts seeking to expand into Python and SQL for data handling
- Aspiring data scientists

Pre-requisites

- Familiarity with Excel or any programming language
- Python (Anaconda or Jupyter) installed
- SQL database setup

Detailed Course Overview and Topics

- **Module 1: SQL Fundamentals:** Master the basics of SQL for querying databases, including SELECT, FROM, WHERE, GROUP BY, and JOIN statements.
- **Module 2: Python Fundamentals:** Learn Python syntax, data types, control flow, and functions.
- **Module 3: Data Analysis with Python:** Dive into data manipulation and analysis using popular Python libraries like Pandas, NumPy, and Matplotlib.
- **Module 4: Advanced SQL & Python:** Learn advanced SQL topics like stored procedures and subqueries, and how to connect Python to a database.
- **Module 5: Capstone Project:** Work on a project that combines your SQL and Python skills to extract, analyze, and visualize data from a real-world database.

On Completion of this Course, You'll Walk Away With:

1

Confidence in **Python** and **SQL** for real-world analytics

2

Ability to **clean**, **analyze**, and **visualize large datasets**

3

A completed coding **project portfolio**

4

Certificate from **Learn Citaville Institute**